

ITA-CondNet

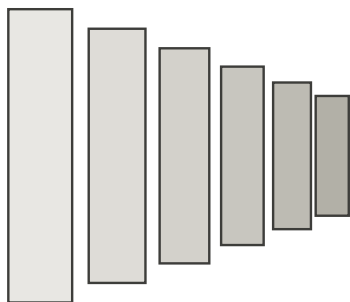
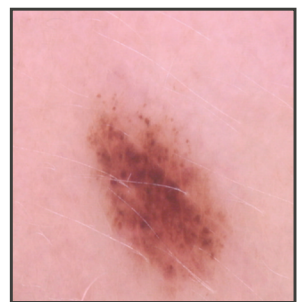
EfficientNet-B2 with an ITA tone branch concatenated at the head

HAM10000 input image

ISIC_0029317 · 224×224

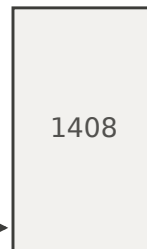
EfficientNet-B2

ImageNet-pretrained

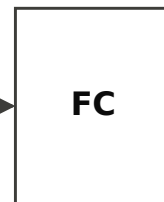


1408-d image features

concat, 1440-d



32



malignancy score

tone conditioning branch



background skin
(Otsu-masked)

$$\bar{L}^* = 77.4 \quad \bar{b}^* = 7.1$$

$$\text{ITA} = \arctan\left(\frac{\bar{L}^* - 50}{\bar{b}^*}\right) \cdot \frac{180}{\pi}$$

$$= 75.5^\circ$$

$$[\sin \theta, \cos \theta, \theta/90]$$

angular encoding, continuous at $\pm 90^\circ$

Linear 3→16→32
+ BN + ReLU

Training AdamW · BCEWithLogits (pos_weight 2.0) · label smoothing 0.05 · CosineAnnealingWarmRestarts · 40 epochs · 8-crop TTA

Five variants Baseline · Aug-only (dark aug + Mixup $\alpha=0.2$ + 3× oversample) · Tone-only · Full · Balanced Baseline (pos_weight 1.0)

Evaluation locked test set n=1,527. Primary analysis on the high-confidence ITA subset: light ITA $> 55^\circ$ (n=707) · dark ITA $< 0^\circ$ (n=558)